

Ideas Register — Consolidated, v1 to v20

One file for every parked idea, with its current status

This file replaces `ideas_v1` through `ideas_v18`. Versions 1–11 were cumulative (each carried the previous forward) and 12–18 were separate additions, so the same idea often appeared in several files at different stages. Here each idea appears once, with where it came from, where it is destined, and where it stands now.

How status was determined. Not from memory of the discussions. Each status was checked by searching the current paper, `Pattern_Arithmetic_v15_22.tex`, and — for items destined elsewhere — the `Assembly_v1.tex` and `Inference_v1.tex` copies on hand (which may be older than your working copies; if so, the “not filed” marks below should be rechecked against yours). Entries from v19 onward are appended here; v19 and v20 are the first two.

Status vocabulary. **CLOSED** — resolved and merged into the paper. **CLOSED (redirected)** — determined not to be the paper’s question; destination named. **RESOLVED, NOT MERGED** — fully worked out but waiting on a prerequisite. **DECIDED** — a ruling recorded; nothing to merge. **PARTLY MERGED** — some of it is in the paper. **OPEN** — still unresolved or unfiled. **PARKED** — verified but with no current home.

1 The gaps of ideas v1, as they now stand

Honest summary first, since this is what prompted the consolidation:

Gap	Status
Gap 1 — Mix across differing directions	CLOSED in v15_21: refused, on the <code>chladni</code> evidence, with Bundle and Collapse remaining available. One small residual (R26 below).
Gap 2 — Location as a sort	RESOLVED, NOT CLOSED. Fully worked out in v17 (meta vector; both encodings verified). It cannot honestly be marked closed in the paper yet: it depends on the meta-vector taxonomy being imported into Paper 1 first, and that taxonomy is not in the paper at all (zero occurrences). The paper’s own “Open: location as a sort” paragraph is therefore still correct as written.
Gap 3 — Reducing a path by merging close actors	CLOSED for Paper 1 in v15_22 (narrowed to the algebraic guarantee); the technique question is redirected to Inference and is not yet filed there.

The two remaining items from v1 are both Inference-destined and neither is filed: the candlestick summary (verified, mechanism works) and the “Inference Mandelbrot” naming flag.

The single most useful next step is importing the meta-vector taxonomy into Paper 1 (R15). It is the prerequisite for closing Gap 2, and it gives a home to location, the two time registers, and V_{derived} at once.

2 Register

ID	Item	Origin	Destination	Status
<i>Closed</i>				
R1	Mix across differing directions (Gap 1)	v1, v3–v4, v7	Paper 1	CLOSED — refused; v15.21
R2	Terminology realignment: Bundle/Remove, Collapse/UnCollapse, Mix/UnMix	v8–v10	Paper 1	CLOSED — merged; v15.20
R3	Symbols $\mathfrak{U}$, $\backslash$, $\oplus$, $\ominus$, $\boxplus$, $\boxminus$	v9–v10	Paper 1	CLOSED — merged ($\ominus$ superseded)
R4	Collapse non-idempotent; N = list length; $A^{(N)}$; UnCollapse algebra	v10	Paper 1	CLOSED — merged
R5	Distinct count D against total N	v13	Paper 1	CLOSED — merged; v15.20
R6	Reducing a path (Gap 3): algebraic guarantee only	v1, v6, v18	Paper 1	CLOSED (redirected) — v15.22
R7	τ_{weather} conflation	v4	—	CLOSED — correction recorded
R8	Same-sort region merge vs cross-sort	v6	Assembly; Paper 1	CLOSED — same-sort solved in Assembly; cross-sort is R1
R9	Internal link, terminology, cohesion audit	v12	Paper 1	CLOSED — fixes in v15.23, carried forward
R10	Stale status lines and symbols in v1–v10	v11	ideas	CLOSED — superseded by this file
R11	Inertia in Assembly via a mass-like weighting	v13	Paper 2	CLOSED — tested; fails
<i>Decided</i>				
R12	Physics names for atom metrics	v13	Paper 1	DECIDED — keep $\mathcal{E}$; N/D earned; “density” stays retired; S/N parked
R13	“uncollapse” (v5) collides with UnCollapse	v11	ideas	DECIDED — call it Mix-provenance link
<i>Resolved but not merged, and its prerequisite</i>				
R14	Gap 2: location and time as meta vectors; registers; age derived	v1, v17	Paper 1	RESOLVED, NOT MERGED — blocked on R15
R15	Meta-vector taxonomy; V_{derived}	v2	Paper 1	OPEN — not in paper; prerequisite for R14
<i>Open — meta vectors</i>				
R16	Link vectors	v2	Paper 2	OPEN — not filed
R17	Meta-vector grouping (promote a shared one)	v2	Inference	OPEN — not filed
R18	Identity handle / Mix-provenance link	v5, v11	Papers 1, 2, Inf.	OPEN — paper has “proposed, not adopted” only
R19	Should Mix/Collapse skip expired contributions?	v17	Paper 1; Inf.	OPEN — deliberately unanswered

ID	Item	Origin	Destination	Status
<i>Open — Collapse</i>				
R20	Collapse across differing sorts (per-vector τ_i)	v10, v11	Paper 1	OPEN — stated in paper; broadened by R36
R21	Collapse as T 's actor	v11	Paper 1	OPEN — “not claimed” in Axiom 7; see R40
R22	Assembly and Collapse: relation	v11	Papers 1, 2	OPEN — Axiom 6 silent
<i>Open — metrics, scope, presentation</i>				
R23	Shell-indexed energy	v14	Paper 1	OPEN — undefined
R24	Antipodal ($\hat{n}$ against $-\hat{n}$) cancellation	v14	Paper 1	OPEN — special case of closed R1
R25	Separable-metric pointwise test	v15	Paper 1	PARKED — verified; value depends on R27
R26	Route-1 scoping clause (refusal is about $\boxplus$ only)	v7	Paper 1	PARTLY MERGED — one sentence not carried
R27	Axiom 7 metric catalogue and costumes: cut or move?	v16	Paper 1	OPEN — undecided
R28	Two-paper split	v16	Papers 1, 2	OPEN — undecided
R29	Review's presentational items	v16	Papers 1, 2	OPEN — not acted on
<i>Open — Inference-bound</i>				
R30	Candlestick summary	v1	Inference	OPEN — verified; not filed
R31	“Inference Mandelbrot” naming flag	v1	Inference	OPEN — flag only
R32	Technique selection for reduction (average, centroids, shape-preserving)	v18	Inference	OPEN — specified; not filed
R33	Douglas–Peucker adapted to the sphere	v18	Inference	OPEN — the one unbuilt piece
R34	Sphere-packing bound for the “stated distance”	v6	Inference	OPEN — not filed anywhere
R35	Coherence check for the candlestick	v1	Inference	OPEN — never considered
<i>Open — atom structure (v19)</i>				
R36	Pure and mixed atoms: cores; per-vector τ_i as the general form; Mona Lisa as test case; direction-based derived quantities per sort	v19	Paper 1	OPEN — broadens R20
R37	Collapse of multi-vector contributions (not fixed at one vector each)	v19	Paper 1	RESOLVED, NOT MERGED — paper still says one vector per contribution
R38	N as number of cores; Q as number of vectors (resolves the collision with N as vector count in Axiom 7)	v19	Paper 1	DECIDED — N cores, Q vectors; paper edits wait for the merge
R39	Collapsed atom as a counted bowl; commuting test	v19	Paper 1	OPEN — test not yet run
R40	T asymmetry: a bowl is not an actor; a Collapsed atom needs a cast	v19	Paper 1	OPEN — relates to R21

ID	Item	Origin	Destination	Status
	<i>Open — energy and effective counts (v20)</i>			
R41	Energy from strength, vector count and direction: $\mathcal{E} = S^2/Q + Q \text{Var}(r)$; $S^2/Q \leq \mathcal{E} \leq S$; $ V ^2 = (1 - \alpha)\mathcal{E} + \alpha S^2$	v20	Paper 1	PARKED — identities verified, not in paper; $\mathcal{E}$ keeps “energy,” $\mathcal{V}$ keeps “volume”
R42	Effective vector count $Q_{\text{eff}} = S^2/\mathcal{E}$; overlap with σ_{atom}	v20	Paper 1	OPEN — candidate; keep it, σ_{atom} , both, or neither
R43	Strength-only metrics per sort and for the whole; effective number of sorts $1/\sum w_s^2$; pooling ratio	v20	Paper 1	OPEN — needs one stated convention

3 The entries, by version

v1 — the three gaps and two Inference items

Gap 1 (Mix across differing directions): Axiom 5 defines Mix only at a shared $\hat{n}$; a provisional vector-sum recipe had been tried and withdrawn. *Status: R1, closed.* **Gap 2** (location as a sort): a third candidate was tested — disjoint dimensions for content, time, and location. Verified: two shifts through disjoint dimensions commute exactly (to full floating-point precision), and content stays recoverable by projection; shifts forced onto one shared dimension do not commute. Left open: the encoding itself. *Status: R14, resolved in v17.* **Gap 3** (reducing a path): needed a distance, a merge rule, and a drift bound. Verified that swapping the clip for the Hill form $r' = |z|/(K+|z|)$ changes the strength but not $\chi' = \arg(z)$. *Status: R6, closed (redirected).* **Candlestick summary:** open/close/high/low distinguishes two sequences of identical content in opposite order that naive Mix cannot — Mix gave the same 47.5° for both; the candle’s open and close swapped ($20^\circ \rightarrow 160^\circ$ against $160^\circ \rightarrow 20^\circ$) while high and low stayed equal. *Status: R30, 0 occurrences in Inference.v1.* **“Inference Mandelbrot”:** an ordered-thing promotion is a different problem from Assembly’s Hill-bijection promotion of unordered camps, and needs its own name. *Status: R31.* The “considered and found to need nothing filed” list (reference atoms, precursor-witness relabeling, the $\pi = (x; w_1, \dots, w_n)$ trajectory, χ as internal phase-color) was rechecked in v11 against the current paper: all present, nothing to file. *Closed.*

v2 — meta vectors, split across three papers

Proposed: besides information vectors (capped, $r_i \leq 1$), an atom may carry meta vectors describing it rather than being part of its content: V_{derived} , link vectors, a timestamp vector. Meta vectors are not capped. Split by destination: the taxonomy and V_{derived} to Paper 1; link vectors to Assembly; the grouping mechanism (query nearby meta vectors, promote one shared value) to Inference. Verified caveat: grouping V_{derived} inherits the Hill form’s limit — both fix magnitude only, never direction. *Status: R15–R17. The paper has zero occurrences of “meta vector”; Assembly.v1 has zero of “link vector.”*

v3 — Gap 1, characterized

Argued reading: Mix takes several reports already agreed to answer one shared question and returns a verdict about it; the shared $\hat{n}$ is what makes that true. Notably, v3 already stated that in Mix’s

domain “ $\hat{n}$ itself was never plural; only χ ever varied” — the very distinction that had to be rediscovered during the Gap 3 discussion (v18) before it stopped causing confusion. Four “we mix all the time” counterexamples were checked and dissolved (sugar in tea is one dial; red plus green is already same- $\hat{n}$; paint compositing is a bowl and flattening is destructive and irreversible — matching UnMix’s impossibility). The one unresolved example, a whole painting as one vector, is Gap 1 itself. The open question split in two: *embedding validity* (does closeness on a shared sphere track semantic agreement?) and the *merge rule*. *Status: superseded by R1’s closure — Mix refuses, so neither sub-question needs an answer for Mix.*

v4 — the τ_{weather} correction; Gap 1 “ready to close”

τ_{weather} had named two different problems: combining different weather concepts across directions (Gap 1), and reducing closely aligned same-sort atoms to one representative, which is Assembly’s already-verified Hill-bijection mechanism (16 camps to 13 systems). The **chladni** section was found to already contain the evidence for Gap 1 (“cross-mode coexistence is therefore $\uplus$ ”). *Status: R7 closed; the chladni evidence is what the Gap 1 closure rests on.*

v5 — identity handle

An opaque *id* meta field, assigned before Bundle or Mix and kept after, so a result can point back to the bowl that produced it (bowl {red, green} is #17; Mix gives yellow #18 with a link #18 $\rightarrow$ #17). Refuse τ_{id} as a sort; keep *id* as a field. Navigation by a kept record, not inversion of Mix. Renamed in v11 “Mix-provenance link” because the original word “uncollapse” collides with the formal UnCollapse operator. *Status: R13 decided; R18 open — the paper mentions id_i once, as “proposed, not adopted,” in UnCollapse’s well-definedness paragraph.*

v6 — sphere-packing bound; merging a whole region

For Gap 3’s “stated distance”: the number of mutually distinguishable points at separation $\theta_{\min}$ is roughly $2/(1 - \cos(\theta_{\min}/2))$ — about 52,525 at 1° , 2,101 at 5° , 526 at 10° , 59 at 30° , 15 at 60° , 7 at 90° , 4 at 120° . Merging a whole region to one representative splits: same-sort is solved (Hill bijection) and cross-sort is Gap 1. *Status: R8 closed; the bound itself is R34, filed nowhere.*

v7 — scoping of Route 1

“Refuse outright” should be worded as a claim about $\boxplus$, not a permanent bar on any future, separately admitted mechanism — the same “undecided, not admitted” logic Axiom 0 already uses. *Status: R26, partly merged.* Checked against the Gap 1 closure: the closure is scoped to Mix and keeps $\uplus$ and $\oplus$ available, which honours the point. What was not carried is the one explicit sentence that a separately admitted mechanism is not barred, and the closure calls the other two routes “withdrawn,” slightly stronger than v7 intended. An optional one-sentence fix.

v8, v9, v10 — terminology and verb inventory

v8 proposed the Collapse/UnCollapse/Bundle/Mix/UnMix realignment and flagged a hard symbol collision ($\ominus$ was Remove). v9 chose $\uplus$ for Bundle and $\ominus$ for UnCollapse, and scoped *ids* to the one case that needs them: duplicate-valued contributions in UnCollapse. v10 revised the symbols — $\ominus$ to UnCollapse, Remove to $\backslash$ — settled Bundle’s idempotence ($A \uplus A = A$, already proven), and showed Collapse is *not* idempotent by three convergent arguments (information loss; an algebraic

contradiction with A UnCollapse $A = 0$; the same evidence-count blindness the Hill form was built to fix). N is simply the length of the vector list; $A^{(N)}$ abbreviates N identical copies; UnCollapse sends $N \rightarrow N - 1$. *Status: R2–R4 closed, all merged. MixMerge was later reverted to Mix at your request; $\ominus$ appears zero times in the paper. The per-vector τ_i question that v10 left open is R20.*

v11 — review of v10 against the paper

Status lines of v8–v10 corrected (the realignment was merged); stale $\oplus/\ominus$ readings in earlier text fixed, including one sentence that had become *false* (“Gather still satisfies $A \oplus A = A$ ” reads wrongly once $\oplus$ means Collapse); the “uncollapse” collision resolved by renaming. Three gaps opened by giving Collapse axiom-level standing: R20 (across sorts), R21 (as an actor for T), R22 (no stated relation to Assembly — Axiom 6 mentions Collapse zero times, so this one is silence rather than a stated omission). One new route recorded: Collapse losslessly first, then define a reading — the seed of what v18 developed. *Status: R10, R13 closed or decided; R20–R22 open.*

v12 — internal audit

96 labels and 198 reference call-sites checked, each against what its sentence claims; zero mismatches. Two stragglers fixed (an Abstract “a Collapse” noun phrase; a Forest-costume “gatherers”). Four cohesion fixes where the paper’s own summaries omitted Collapse, and the UnMix proof section now names UnMix. No new conceptual gap. *Status: R9 closed; fixes are in the current paper.*

v13 — naming the atom metrics

Seven metrics, quoted precisely. Earlier drafts had called $\mathcal{V}_{\text{atom}}$ “mass,” γ_{atom} “density,” and S_{atom} “atom energy,” and retired all three; only $\mathcal{E}_{\text{atom}} = \sum r_i^2$ kept “energy,” because it is tied by an exact identity to $|V_{\text{derived}}|^2 = \mathcal{E}_{\text{atom}} + 2 \sum_{i < j} r_i r_j (\hat{n}_i \cdot \hat{n}_j)$ (variance-of-a-sum structure, verified to six decimals). The rule: a physics name is kept only if the quantity participates in something provable. $D = |A_1 \uplus \dots \uplus A_n|$ passes (forced by $\uplus$ ’s idempotence), so N and D keep the mass-number/atomic-number analogy; “density” stays retired; S_{atom}/N is parked rather than named. Inertia in Assembly is closed: a mass-like weighting was already tested in the motion equation — a ninefold strength gap gives a 729-fold weight and a non-monotonic trajectory. *Added in conversation after v15, not previously filed:* asked whether either of the two metrics survives, the answer was that energy already does ($\mathcal{E}_{\text{atom}}$, earned) while no candidate for density does — S_{atom} already has a name, N is a count rather than a ratio, and the one ratio with the right shape, γ_{atom} , is the one already rejected. *Status: R5 closed; R11 closed; R12 decided.*

v14 — three rough sketches

Density restated: nothing new, closed by reference to v13. *Energy as shell distribution:* partly already true of $\mathcal{E}_{\text{atom}}$ — for fixed total strength $S = 2.0$, four vectors give $\mathcal{E} = 1.0$ spread evenly and 1.81 concentrated toward the extremes (and σ_{atom} goes $0 \rightarrow 0.333$) — but σ_{atom} is blind to *which side* the concentration is on, and no shell-indexed quantity exists. *Matter/antimatter:* antipodal cancellation ($\hat{n}$ against $-\hat{n}$), a second kind of opposition beside the existing same-direction opposite-phase cancellation. With Gap 1 now closed by refusal, this would be a deliberate special-case exception, parallel to the existing one; what the rule would produce, and whether it needs equal strength, is undecided. *Status: R23, R24 open.*

v15 — when two metrics constrain each other

For separable metrics $M_1 = \sum f(r_i)$ and $M_2 = \sum g(r_i)$: if $f \leq g$ everywhere on $[0, 1]$ then $M_1 \leq M_2$ for every atom; if the curves cross there is no bound. Verified against both cases (it forces $\mathcal{E} \leq S$ because $r^2 \leq r$; it forces nothing for N against a shell-weighted energy). Where a bound holds, the gap is itself a metric ($r - r^2$, largest at $r = 0.5$). Possible uses (a non-redundant reporting set, a two-axis placement, a debugging check, a post-hoc label for Assembly’s camps) are speculative. *Status: R25, parked; it matters mainly if R27 keeps the metrics.*

v16 — external review and the two-paper split

Every checkable claim in the review was verified: the worked numbers (e.g. $r' \approx 0.721$ at 73.9° ; freeze at 628 and 1261 steps), that all 22 occurrences of the six Axiom 7 metrics sit inside Axiom 7 and none in Bundle/Mix/T/offset, that “white” is used in two senses (the paper already says so in Axiom 3), and that the “checked directly” precision claims have no supplement. The proposal: Paper 1 is the typed algebra, Paper 2 is assembly and the enclosure, the Axiom 7 catalogue and the costumes go in neither; Paper 1 states Assembly as a signature only and ends by naming the seat it refuses to supply; cut by question, not by length. *Not itemized when v16 was filed, but in the review and still pending (R29):* state the clip, the Mix refusal, and the assembly energy as stipulated choices in one place rather than narrated as discoveries; print the enclosure gradient; provide a supplement for the “checked directly” claims; and stop calling the enclosure a “law” in the same voice as $SO(3)$ while it still needs δ , η , the census cut, and $\varepsilon_{\text{allow}}$. *Status: R27, R28, R29 open and undecided.*

v17 — Gap 2 resolved

Meta vector, not sort, tested against the cap: $r \leq 1$ encodes strength of content, while location and time are coordinates. This is the line the paper already draws for Assembly’s seat. Location is one vector via standard (θ, ϕ) from latitude and longitude (London $\rightarrow (0.6225, -0.0011, 0.7826)$, Tokyo $\rightarrow (-0.6194, 0.5252, 0.5835)$, Sydney $\rightarrow (-0.7273, 0.3999, -0.5577)$). Time on (θ, ϕ) fails: at hour 0 every minute lands on $(0, 0, 1)$, and 11:59 and 0:00 land about 2 apart (antipodal). The fix: minute on ϕ , hour on χ , θ fixed at the equator (6° and under 1° apart for the same pair). Timestamp is a register of length N , kept in step with UnCollapse; age is derived (now – timestamp), never stored; expiry is a second register because it cannot be derived; timestamps help but do not guarantee disambiguation the way an *id* would. **Blocking:** cannot be merged until R15 is. *Status: R14, R19.*

v18 — Gap 3 closed for Paper 1

Corrected an earlier confusion (χ is phase at a shared direction, not $\hat{n}$; Gap 1’s refusal did not apply). The real issue is order-blindness: merging three actors in either order gives the same result. A path’s commitment to order comes from outside the algebra — Bundle discards order, and choosing T over a bowl is the declaration that order matters. So averaging 10,000 temperature readings is bowl-plus-Mix and loses nothing. Three techniques answer three different questions: averaging cannot tell a spike from a flat line; centroids give identical results for a sequence and its reverse (clustering is blind to position); shape-preserving reduction keeps real points at their positions — Douglas–Peucker, whose tolerance is the drift bound Gap 3 asked for. Paper 1 was narrowed to the non-consumption guarantee. *Status: R6 closed (redirected); R32, R33 open for Inference.*

v19 — pure and mixed atoms, cores, and Collapse against a bowl

Prompted by a hypothetical: could the Mona Lisa be held in one atom? “One atom” can mean three things. *A name*: a single seated point, like “storm,” which the paper calls a doorbell — it carries none of the content. *A bowl*: already legal for mixed sorts, since Bundle is sort-blind. *A Collapsed atom* holding colors, words, and strokes, which needs mixed sorts inside one atom. This is the example v3 left unresolved (“a whole painting as a single vector” was Gap 1 restated). With Gap 1 closed, a single *vector* is refused, while one atom of many vectors stays available. What stands in the way of the third meaning: mixed sorts (R20, widened to R36); where each contribution sits on the canvas — a location per contribution, the same kind of register as v17’s but not yet specified per contribution; derived quantities across sorts; and relations, since the paper “does not mint the relations” and, by its own design rule, which units were co-present and within what region are facts of a level, not coordinates of an atom.

Rulings given in the discussion, none yet in the paper.

- *Collapse juxtaposes.* Positions do not change, nothing merges, no new type is minted. Relations within a sort are as before; across sorts they exist only under a stated rule (Axiom 0’s “undecided, not admitted,” applied to Collapse). So derived quantities that use direction are computed per sort, and cross-sort ones are undefined until a rule exists. (Refined in v20: strength-only quantities may also be given for the whole atom.)
- *Cores.* A core is a unit inside an atom, usually one τ (a pure type) but free to hold several — words and numbers together — as typical, not as a rule. The general form is per-vector τ_i , with “core” a grouping. “Core” appears in the paper only in the “Core Contributions” heading and in “no core, no halo” in the enclosure, so it needs one defined sense.
- *Collapse is not fixed at one vector per contribution.* This settles the question the paper was silent on: whether collapsing already-multi-vector atoms flattens or nests. The Collapse text still says “one vector per contribution.”
- *N is the number of cores.* This keeps UnCollapse’s $N \rightarrow N - 1$, keeps D a bowl size (distinct cores), and keeps the v17 registers at length N , so those registers are per core, not per vector. The cost is a word collision: N is currently the vector count in Axiom 7, in $\gamma_{\text{atom}} = \mathcal{V}_{\text{atom}}/N$ and in σ_{atom} ’s $N^2/(N - 1)$. *Decided:* N counts cores and Q counts vectors. All seven Axiom 7 metrics run over vectors, so they take Q ; the D/N paragraph and the UnCollapse text keep N and change “vectors” to “cores.” To settle at the merge: the sentence that defines N as the atom’s count of informational vectors, and where the label “info number” goes.
- *A Collapsed atom matches a counted bowl.* They are identical exactly when nothing repeats ($D = N$), and $N - D$ is the gap. The paper’s bowl is a set, and its own counting equation shows the difference: $A \uplus A \uplus A \uplus A \setminus A = \emptyset$, while $A \oplus A \oplus A \oplus A \ominus A = A^{(3)}$ (by the UnCollapse rule). Operations should work on both; recorded as a commuting test — a pack/unpack map between the two, with every operation required to agree across it. Sum-style quantities (S , $\mathcal{E}$, V_{derived}) agree only on a counted bowl. T is asymmetric today: it “does not act on the set R as a set,” and a Collapsed atom can act only once a cast is named.
- *Two distinct ways of arranging and viewing vectors.* They differ in multiplicity (set against counted), in handle (a Collapsed atom is one unit that can sit in a bowl or at a lexicon pin; a bowl is a collection), and in sequence (neither stores order; the path carries it, and a bowl is a good surface for displaying one).

Status: R36–R40; R20 and R21 cross-referenced.

v20 — energy from strength, counts, and direction; effective counts

Started from a recollection that energy is $\sum r^3$. The paper’s energy is $\mathcal{E} = \sum r^2$; $\sum r^3$ is the atom’s volume, formerly “mass.” Nothing needs renaming, and $\sum r^3$ keeps a geometric reading (strength as a radius, as in Assembly’s softening length).

Energy from strength, vector count Q , and direction already exists in $\mathcal{E}$. Three exact relations, verified on 20,000 random atoms (errors near 10^{-14} , no violations), with $S = \sum r_i$:

- $\mathcal{E} = S^2/Q + Q \text{Var}(r)$: an evenly spread floor, set by S and Q alone, plus concentration.
- $S^2/Q \leq \mathcal{E} \leq S$.

- $|V_{\text{derived}}|^2 = (1 - \alpha)\mathcal{E} + \alpha S^2$: the resultant’s energy, blending the incoherent baseline with the fully aligned value, weighted by the paper’s coherence α .

Which works better: the floor S^2/Q cannot tell an even atom from a concentrated one (both give 1.000 at $S = 2$, $Q = 4$, where $\mathcal{E}$ gives 1.000 against 1.810). $\mathcal{V} = \sum r^3$ separates them even more sharply (0.500 against 1.715) but has no identity tying it to anything beyond the radius reading. So $\mathcal{E}$ is the energy and $\mathcal{V}$ stays volume. Cores N enter through the same variance-of-a-sum algebra applied per core (plain algebra, not run separately) and through the ratios Q/N and S/N . The mean strength S/Q , the parked candidate from v13 (then written S/N), is the centre of the variance decomposition and keeps a plain name.

Effective vector count $Q_{\text{eff}} = S^2/\mathcal{E}$ (the participation ratio in physics, the effective sample size in statistics).

- Q counts vectors and Q_{eff} counts those that matter: 100 vectors at 0.05 give 100; one vector at 0.9 plus 99 at 0.001 give 1.23.
- The largest vector carries at least $1/Q_{\text{eff}}$ of the strength (50,000 random atoms, no violations).
- Scale-free: halving every r leaves it unchanged (2.2099 both).
- With direction: $|V|^2/S^2 = \alpha + (1 - \alpha)/Q_{\text{eff}}$ (error near 7×10^{-16}), so two dimensionless numbers fix the resultant’s size.
- Limits: it re-expresses S and $\mathcal{E}$, adding no information, and ignores direction.
- Overlap: $\sigma_{\text{atom}} = (Q/Q_{\text{eff},\mathcal{V}} - 1)/(Q - 1)$, where $Q_{\text{eff},\mathcal{V}}$ is the effective count of the *cubed* shares (error near 10^{-15}). So σ_{atom} is already this family, and Q_{eff} is its plain-strength sibling, the one tied by identity to $\mathcal{E}$ and $|V|^2$. Whether to keep both, one, or neither is undecided.

Per sort and for the whole. The per-sort rule belongs to quantities that use direction (V_{derived} , α , $|V|^2$), since sums and dot products of directions across sorts mean nothing. Strength-only quantities (S , $\mathcal{E}$, $\mathcal{V}$, Q_{eff} , the σ ’s) can be given per sort and for the whole, on one stated convention: each sort’s r is normalized to its own declared saturation (the paper’s $r = a/a_{\text{max}}$), and whole-atom figures treat fraction of own saturation as a common scale. With w_s the share of total strength held by sort s :

- Pooling ratio $\sum_s Q_{\text{eff},s}/Q_{\text{eff,whole}} \geq 1$ (Cauchy–Schwarz; no violations in 30,000 atoms), with equality exactly when every sort has the same strength-weighted mean strength $\mathcal{E}_s/S_s$. It measures imbalance between sorts: two vectors at 0.9 against ten at 0.1 give 2.63.
- Exact decomposition: $1/Q_{\text{eff,whole}} = \sum_s w_s^2/Q_{\text{eff},s}$ (error near 2×10^{-16}). In the symmetric case it multiplies: three sorts of four vectors at 0.5 give $12 = 3 \times 4$.
- Effective number of sorts $1/\sum_s w_s^2$ answers “is this atom really mixed?” more readably than the ratio: for 50 color vectors at 0.5, 3 word vectors at 0.5, and 10 stroke vectors at 0.3, Q_{eff} is 50, 3, 10 per sort, 61.5 for the whole, and the effective number of sorts is 1.37. Three sorts are present but the atom reads as about 1.4, because color carries 85% of the strength.

Status: R41–R43; the v19 per-sort line is refined above.

4 Merged directly, without an ideas entry

These went straight into the paper at your direction and are already in `Pattern.Arithmetic.v15-22.tex`: the Pāṇini expansion (pratyāhāra and vipratīṣedha as structural parallels, with the Ingerman 1967 attribution of “Pāṇini–Backus form” and the oversimplification caveat); the paragraph on summing separate realities; the quantum-mechanics/general-relativity paragraph, deliberately hedged so that it does not claim Pattern Arithmetic bridges them; and the N/D addition (filed afterwards as R5).

5 What remains, in dependency order

1. **Import the meta-vector taxonomy into Paper 1 (R15).** It unblocks R14, which closes Gap 2 and gives location, the timestamp and expiry registers, and V_{derived} a home.
2. **Then merge the v17 resolution (R14),** replacing the paper’s “Open: location as a sort” paragraph. Decide R19 (expired contributions) at the same time or defer it explicitly. If R38 is adopted, v17’s “per vector” registers should read “per core.”
3. **Decide the scope questions (R27, R28):** Axiom 7’s metrics and the costumes, and whether to split. These also decide whether R12, R25, and R41–R43 still matter for Paper 1.
4. **The atom’s structure (R20–R22, R36–R40).** The word is settled (R38: N cores, Q vectors). Decide the general atom form — cores, per-vector τ_i , contribution size free (R36, R37) — which closes R20 in the same step. R21 and R22 can each be closed by a sentence, whether a stated refusal or a stated rule, and the commuting test (R39, R40) can then be run.
5. **Package the Inference-bound items (R17, R30–R35)** as one coherent addition to the Inference paper: the candlestick, the technique-selection argument, the spherical Douglas–Peucker adaptation, the sphere-packing distance, and meta-vector grouping.
6. **Small residuals:** the one-sentence R26 clause, the review’s presentational items (R29), shell-indexed energy (R23), the antipodal special case (R24), and whether S_{atom}/N earns a name.